

FYP22 Monthly Progress Report: September 2026

Group: 27	Project: Beyond Optical PPG: mmWave Radar Arterial Pulse Sensing
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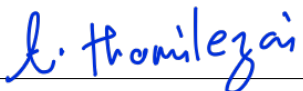
Overall Monthly Summary

This month our work moved from component-level hardware selection into full schematic design of the custom wearable PCB in Altium, alongside continued data analysis and physical prototyping of the wrist-worn form factor. Kopithan completed the schematic sections for the radar, power supply, microcontroller, flash memory and connectors, following Infineon's hardware design guidelines for the BGT60TR13C, including a separate ferrite bead and decoupling capacitors per supply pin, the 80 MHz reference oscillator, and the SPI, reset and interrupt connections to the STM32H743 microcontroller. He also connected the microcontroller's crystal oscillator, boot switch and SWD programming header, added a NAND flash on its own SPI bus, and a USB-C connector with a power path switching between USB and battery supply. A key open issue was identified in the power supply design: Infineon requires any switching regulator used for the radar to run at a fixed frequency of at least 1.5 MHz without dropping during operation, to keep switching noise out of the radar IF band. This has been raised with the team for further discussion before the regulator arrangement and the 1.8 V/3.3 V power connections are finalised.

Mathujan worked on the schematic design of the IMU and PPG sections. For the IMU, he integrated the Bosch BMI270 over I2C, configuring the address and interrupt pins according to the recommended setup and providing decoupled 3.3 V supplies. For the PPG section, he worked with the MAX86174 optical AFE and an SFH 2270R photodiode for green-wavelength detection, submitting an NDA request through the Analog Devices support portal to access the full datasheet for our academic research use, and implemented I2C configuration and dedicated grounding for the optical sensor to reduce noise. He also designed a dual-stage power regulation approach using an AP2112K LDO for the 3.3 V rail and a TPS62840DLCR buck converter with a shielded inductor for the 1.8 V rail feeding the AFE.

Ananthakumar focused on analysing the radar data collected so far, verifying the close-range readings to confirm the sensor was capturing usable signal at the short standoff distances relevant to our wrist-worn setup, and was involved in wrist strap selection and further data collection sessions. She also updated the ethical review documentation to reflect the group's revised project scope and experimental setup, keeping the participant-facing descriptions of the sensing devices and procedures consistent with the current design.

Siva worked on the wrist strap selection in parallel, evaluating strap materials and mounting options that would hold the radar and PPG sensors consistently against the skin without adding excessive bulk, and contacted Infineon directly to clarify the BGT60TR13C's minimum detectable distance so this constraint could inform both the strap and enclosure design. He worked with Ananthakumar on reducing the bulkiness of the enclosure now that the radar, PPG sensor, IMU and battery all need to be accommodated on the wrist, and with Kopithan on the enclosure layout itself, arranging components more compactly while keeping the radar's field of view and minimum standoff distance unobstructed. He also continued refining the contextual bandit formulation for the adaptive algorithm, to be carried forward once the custom PCB work is complete.

Index no.: 220029T	Name: Ananthakumar T.
Detailed list of work carried out (including problems encountered and solutions). This month I worked on analysing the radar data collected so far, focusing on verifying the close-range readings to check that the sensor was correctly capturing usable signal at the short standoff distances relevant to our wrist-worn setup. I was also involved in the wrist strap selection, working with the group to evaluate options that would keep the radar and PPG sensors held consistently against the skin, and took part in further data collection sessions to build on our initial dataset. Alongside this, I updated the ethical review documentation to reflect the changes needed following the group's revised project scope and experimental setup, making sure the descriptions of the sensing devices and procedures given to participants remained accurate and consistent with the current design.	
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Index no.: 220327F	Name: Kopithan M.
Detailed list of work carried out (including problems encountered and solutions). During November, my main task was to complete the schematic design of the wearable device in Altium. I mainly worked on the radar, power supply, microcontroller, flash memory and connector sections, and I also contributed to some parts of the PPG and IMU sections. For the BGT60TR13C radar, I followed the Infineon hardware design guidelines and added a separate ferrite bead and decoupling capacitors for each supply pin, the 80 MHz reference oscillator with its series resistor, and the SPI, reset and interrupt connections to the microcontroller. For the microcontroller section, I connected the STM32H743 with its decoupling capacitors, crystal oscillator, boot switch and SWD programming header, and assigned the pins for the radar, PPG, IMU and flash buses. I added the NAND flash on its own SPI bus, and a USB-C connector with CC resistors and a simple power path that switches between the USB and battery supply. The main problem I found was in the power supply. Infineon recommends that any switching regulator used for the radar must run at a fixed frequency of at least 1.5 MHz, and this frequency must not drop while the radar is operating. Otherwise, the switching noise can fall inside the radar IF band. I have not changed the power circuit yet, because the best regulator arrangement still needs to be decided. Instead, I raised this issue with the team and took the initiative to start further discussions before finalising it. The connections between the 1.8 V and 3.3 V parts are also still under review, and some components may change. Finally, I checked the component values and package sizes around the radar section against the datasheet limits. Once the power supply decision is finalised, the schematic will be ready for the PCB layout stage.	
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Index no.: 220389U	Name: Mathujan S.
Detailed list of work carried out (including problems encountered and solutions). During November, I mainly worked on the schematic design of the IMU and PPG sections of our wearable device in Altium. I also contributed to the selection and configuration of the external flash memory and power supply components required for the custom PCB. For the IMU section, I integrated the Bosch BMI270 sensor and configured it for I2C communication. I set the I2C address to 0x68 by connecting the SA0 pin to ground and configured the CSB pin for I2C mode. The INT1 pin was connected to an external interrupt pin of the microcontroller, while unused INT2 and OIS pins were left open according to the recommended configuration. I also provided clean 3.3 V supplies for VDD and VDDIO with the required 100 nF decoupling capacitors. I also worked on the PPG section using the MAX86174 optical AFE and the SFH 2270R PIN photo-diode for green-wavelength detection. Since the complete MAX86174 datasheet requires an NDA, I submitted an NDA request through the Analog Devices support portal for our non commercial academic research and prototype development. In the schematic, I configured the AFE for I2C communication, added the required decoupling capacitors, and implemented dedicated grounding for the optical sensor section to reduce noise. For the power supply, I designed a dual-stage regulation approach using the AP2112K LDO to generate the 3.3 V rail and the TPS62840DLCR buck converter to generate the 1.8 V supply required by the AFE core and digital interface. A shielded inductor was selected for the buck converter to help reduce the possibility of switching noise	
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Index no.: 220619D	Name: Sivamynthan N.
Detailed list of work carried out (including problems encountered and solutions). This month I worked on the wrist strap selection for the prototype, comparing available strap materials and mounting options to find one that would hold the radar and PPG sensors firmly against the skin without adding excessive bulk. Since our adaptive algorithm and overall sensing accuracy depend on the radar maintaining a consistent, short standoff distance from the skin, I contacted Infineon directly to clarify the minimum detectable distance for the BGT60TR13C, so that this constraint could be factored into both the strap and enclosure design rather than discovered later during testing. Together with Ananthakumar, I worked on addressing the bulkiness of the enclosure, which had become a concern once the radar, PPG sensor, IMU and battery were all accounted for on the wrist-worn form factor. I also contributed to part of the enclosure design together with Kopithan, working through how the components could be arranged more compactly while keeping the radar's field of view and minimum standoff distance unobstructed. On the algorithmic side, I continued developing the adaptive radar approach and gained further insight into refining the contextual bandit formulation, which I plan to carry forward into the next phase once the custom PCB work begins.	
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Supervisor's Certification

Supervisor's Name Dr.Sampath Perera	Signature	Date
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